PROJECT 8C / ENGINEERING MASTER BRIEF

COMPLETE CHASSIS CONSTRAINTS DOSSIER

For Codex-driven, fabrication-oriented FreeCAD chassis redesign

BODY BASIS	1938 Alfa Romeo 8C 2900B Lungo Spider; treat 1938/1939 geometry as equivalent for this project.
ENLARGEMENT	Uniform 2100/1770 = 1.18644068 scaling of provisional historic reference values. No deliberate styling restyling.
MODULES	01 Front / GX100 suspended front; 02 Cabin / two-seat roadster; 03 Rear / Toyota Probox 1NZ-FE plus transaxle study.
DESIGN STATUS	DESIGN INPUTS ONLY. No donor mounting coordinate, weld specification, fatigue result, axle loads, or legal approval is verified.

Codex mission: produce a clean, coherent and maintainable structural chassis, not an indiscriminate union of prior STEP concepts. Preserve approved geometry; parameterise, reconcile and independently check everything else.

NON-NEGOTIABLE	No fabrication release, no machining hole pattern, no suspension bracket welding, and no road use merely because CAD geometry is valid.

Prepared 11 October 2026 | C17 Rev 0 | Source types: project-generated concepts, project planning notes and independently catalogued Toyota part families.

01 How Codex must interpret this dossier

Authority order and configuration lock

- Priority A: physically measured donor hardware, certified/approved engineer requirements and finalized body/occupant package - when later supplied. These override concept CAD only through explicit logged revisions.
- Priority B: user-approved scaling policy, 1938/1939 equivalence assumption, 3,559 mm nominal wheelbase, master axes, and selected donor families.
- Priority C: C16 GX100 front-mount concept supersedes floating C15 markers; C11 cabin concept is a spatial template only; C09 rear is envelope study only.
- Priority D: speculative mount coordinates, brackets, section sizes, drivetrain packaging and tyre envelope are editable study constraints and never treated as measured hardpoints.

Meaning of constraint statuses

Code	Meaning	Codex behaviour
F / PROJECT FIXED	Current agreed project target (not necessarily a certified historical fact).	Keep fixed unless user approves a new revision; compare to future measurements.
D / DERIVED	Calculated directly from a fixed project reference.	Recalculate using one master expression, do not hand-key many rounded versions.
P / PROVISIONAL	Geometric concept, packaging guess or research estimate.	Keep parameterised; tag concept; allow revision; never drive a purchase/cut file.
M / MEASURE	Requires a physical scan/donor inspection or identified OEM drawing.	No guessed drilling/critical interface values. A labelled proxy may be shown.
E / ENGINEER	Requires independent structural, regulatory or safety design.	Hold release until load cases, calculations, review and checks complete.
X / CONFLICT	Two current project inputs cannot both be relied upon as final.	Emit explicit constraint failure and an options/impact report before committing.

Core task rules

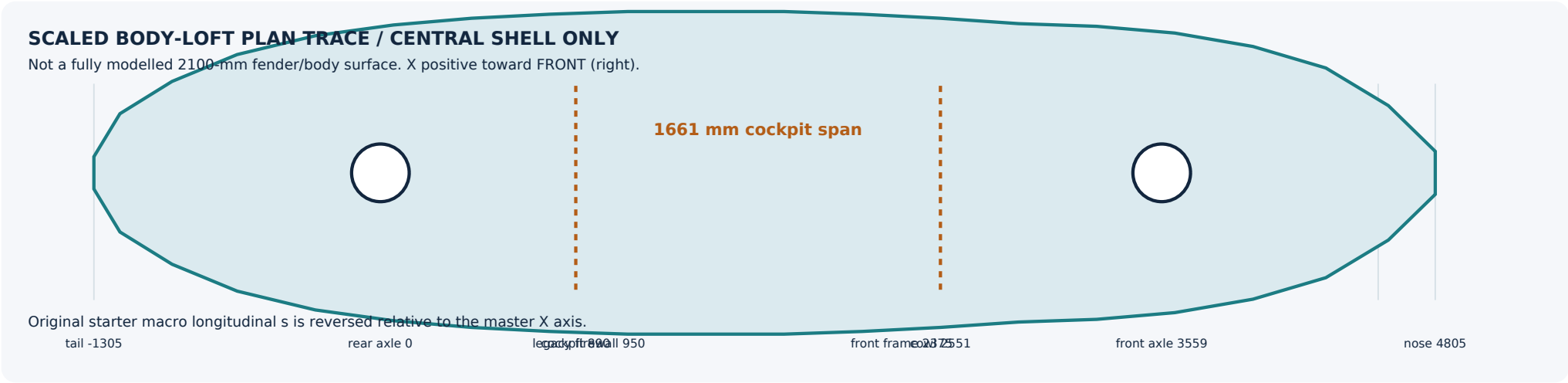
- Keep donor dimensions 1:1. Only scale the chosen classic reference shape; do not multiply engine, transaxle, hub, bearing, link, body-mount, bolt, or tyre dimensions by 1.18644.
- Replace trial features with measured features without changing the origin, handedness or naming; preserve provenance per geometry item.
- When data is missing, generate an envelope and a requirement for measurement. Never silently synthesize an OEM-compatible bracket or claim an FEA-pass.

02 Master datums, body scale and coordinate invariants

Parameter / expression	Value (mm unless stated)	Status / interpretation
SCALE = 2100 / 1770	1.18644068	F - uniform scaling of provisional historical reference body
Ref length: 5150*S	6110.17 (~6110)	D - body overall target; original 5150 reference not independently validated
Ref max width: 1770*S	2100.00	F - overall BODY/FENDER design target
Ref max height: 1500*S	1779.66 (~1780)	D - height datum still requires silhouette definition
Wheelbase: 3000*S	3559.32 (~3559)	D - wheel centre stations; project nominal
Vehicle X axis	+ forward	F - global master coordinate
Vehicle Y axis	+ vehicle LEFT	F - right is negative Y
Vehicle Z axis	+ up	F - ground plane Z = 0, provisional loaded attitude
Rear wheel station	X = 0	F - global origin longitudinal station
Front wheel station	X = +3559.32	D - public-facing drawings may use +3559
Body macro reversal	X = (4050 - s) * S	D - s is original starter macro station increasing toward rear
Frontmost body tip	X = +4805.08	D - from s=0
Rearmost body tip	X = -1305.08	D - from s=5150
Front axle height (tyre study)	Z = +330	P - for 660 OD trial front tyre, not a fixed body datum

AXIS TRAP The body-starter macro uses original s positions from bonnet (s=0) toward tail (s=5150). It is opposite the current chassis master X direction. Always transform s using (4050 - s)*S; never import s directly as the chassis X coordinate.

03 Scaled silhouette interpretation / actual body data



The C10 cabin dimension study is based on the starter FreeCAD central shell loft. It does not yet define the actual separate fender surfaces or a fully constrained 2,100-mm vehicle exterior. Do not invent the fender sweep to force the rail fit.

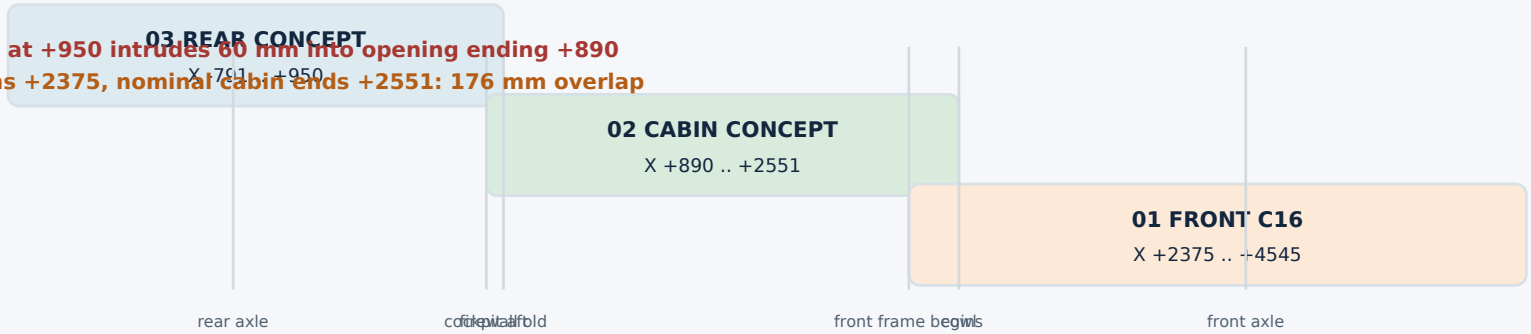
Station checkpoints used by the chassis team

Chassis X	Nominal central-shell width	Bottom Z	Shoulder Z	Crown Z	Evidence
+2550.85	1364.41	308.47	937.29	1245.76	starter station s=1900
+2194.92	1400.00	308.47	972.88	1328.81	s=2200
+1838.98	1423.73	308.47	984.75	1352.54	s=2500
+1483.05	1423.73	308.47	984.75	1352.54	s=2800
+1127.12	1423.73	308.47	972.88	1316.95	s=3100
+889.83	1407.91	308.47	957.06	1261.58	interpolated C10
+771.19	1400.00	308.47	949.15	1233.90	s=3400

04 Three structural sections and their conflicts

THREE DESIGN PACKAGES / NOT THREE AUTONOMOUS LOAD PATHS

REAR: old firewall at +950 intrudes 60 mm into opening ending +890
FRONT: C16 begins +2375, nominal cabin ends +2551: 176 mm overlap



Domain and integration contract

Module	Existing span (mm)	Do not assume
01 FRONT - C16	X +2375 .. +4545	A validated cowl flange at +2551; a road-rated suspension geometry; safe nose crush zone.
02 CABIN - C11	X +890 .. +2551	A fully enclosed cockpit floor, an engineered side intrusion system, a firewall or true vehicle belt mounts.
03 REAR - C09/C08	X -791 .. +950	OEM Probox mounting coordinates, safe rear crash zone or proven rear transaxle installation.

CONFLICT X-01 / REAR	Cockpit opening aft edge X +889.83; historical rear firewall plane X +950. The firewall projects 60.17 mm into the cockpit opening. Do not move or freeze the firewall until seat/backrest, thermal bulkhead, rear engine and service packages are solved.
INTERFACE X-02 / FRONT	C16 front frame begins X +2375 while C11 ends X +2551. Their 176 mm overlap is a co-design lap zone, not proof that the CAD solids match or that a weld/bolted joint works.
PACKAGING X-03 / WHEELS	Uniformly scaled body wheel-reference radius from original 330 mm becomes ~391.5 mm; C14 front tyre radius is 330 mm. The scaled starter arch radius (330+45)*S becomes ~444.9 mm. These do not correspond to the same wheel size/centres; preserve styling, resolve actual tyres and fender sweep separately.

05 Structural architecture / clean redesign brief

Preferred concept for refinement

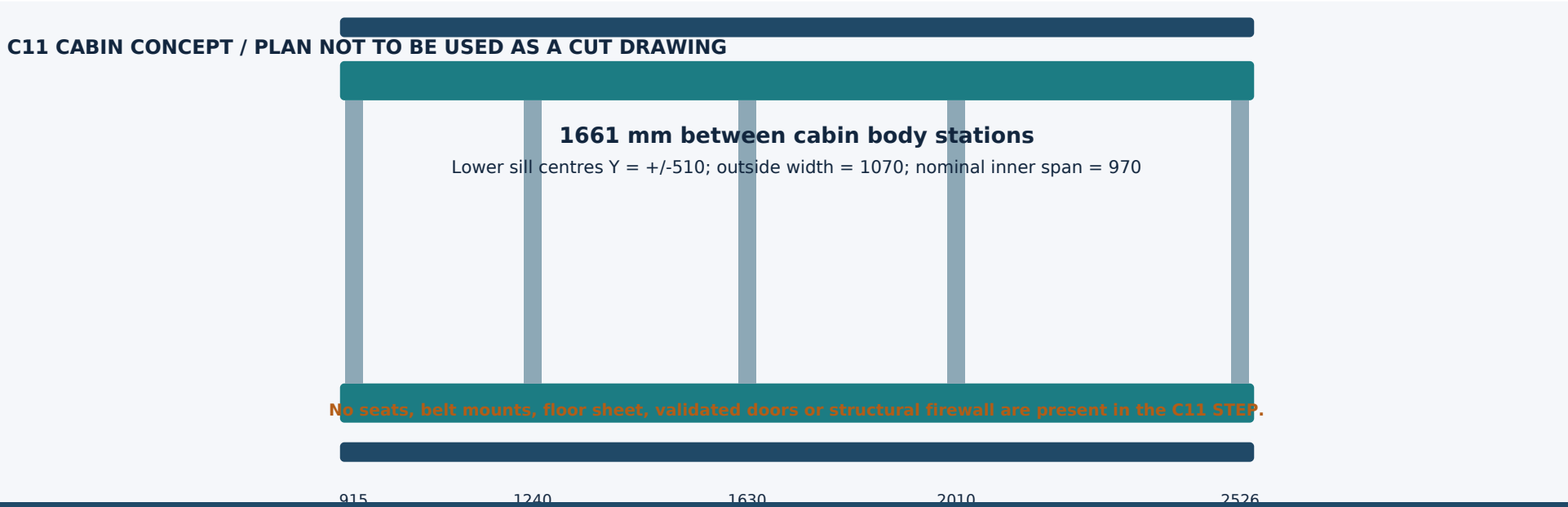
Develop an integrated, welded RHS chassis with two continuous longitudinal load paths, paired deep side sill systems around the open occupant cell, triangulated front suspension cradles, and a replaceable but structurally supported rear donor cradle where engineering permits. Use the C11, C16 and rear CAD as reference studies, not mandatory member-to-member geometry.

Subsystem	Design intent	Evidence for acceptance
Continuous primary rails	Carry front wheel/braking loads through cabin into rear module; avoid discontinuous unsupported splice stubs.	Section force paths, load transfer diagrams, joint accessibility and FEA.
Convertible side sills	Torsion-efficient deep closed/multi-cell box sill subject to door geometry and ingress.	Open-roof torsion, side loads, occupant and panel clearance.
Floor diaphragm	Shear-capable crossmembers and optional engineered metal shear panels; removable service panels not assumed structural.	Shear flow, weld/bond/fastener detail, seat load integration.
Front suspension support	Independent GX100 upper/lower pivot loads + coilover loads integrated into dual planes; no floating brackets.	Actual OEM interface scans, static loads, fatigue and bump steer.
Rear cradle	Probox subframe/engine/transaxle loads into rear bays; isolate heat and service from occupants.	Measured subframe bolt nodes, powertrain orientation feasibility, cooling and rear impact.
Bulkhead / rollover	Engine/fire boundary and separately designed belt/occupant protection paths.	Fire barrier, seat/belt load calculations, roofless-vehicle safety plan.
Body supports	Nonprimary composite mounts with replaceable rubber/isolation where suitable.	Measured surfaces, no bracket piercing a required clear zone.

Geometry hygiene - required for each member

- Member defined by a named start node/end node, cross-section outer size, wall thickness, steel grade, orientation vector, end cut description, and attachment group; no anonymous floating tubes.
- Single shell/beam load path first, then solid RHS with manufacturable joints. Avoid hollow member end collisions, inaccessible welds, impossible sleeve installation and abrupt thickness changes.
- Use mirrored geometry only after verifying true LH/RH Toyota asymmetry and toe/steering geometry; mounting holes and nuts must have tool access.

06 Cabin C11 dimensional register / occupant zone



C11 member group	Geometry (trial)	Current location / status
Lower sill pair	RHS 100 x 50 x 3, vertical height 100	Y +/-510 centre; X +890..+2551; Z 445..545 / P
Upper sill pair	RHS 60 x 40 x 3, vertical height 60	Y +/-510 centre; X +890..+2551; Z 675..735 / P
Side connectors x10	RHS 40 x 40 x 3, length 130	Both sides at X +915,+1240,+1630,+2010,+2526; Z545..675 / P
Floor crossmembers x5	RHS 100 x 50 x 3, span 970	X +915,+1240,+1630,+2010,+2526; Y -485..+485; Z445..545 / P
Upper end ties x2	RHS 60 x 40 x 3, span 980	X +915 and +2526; Y -490..+490; Z675..735 / P

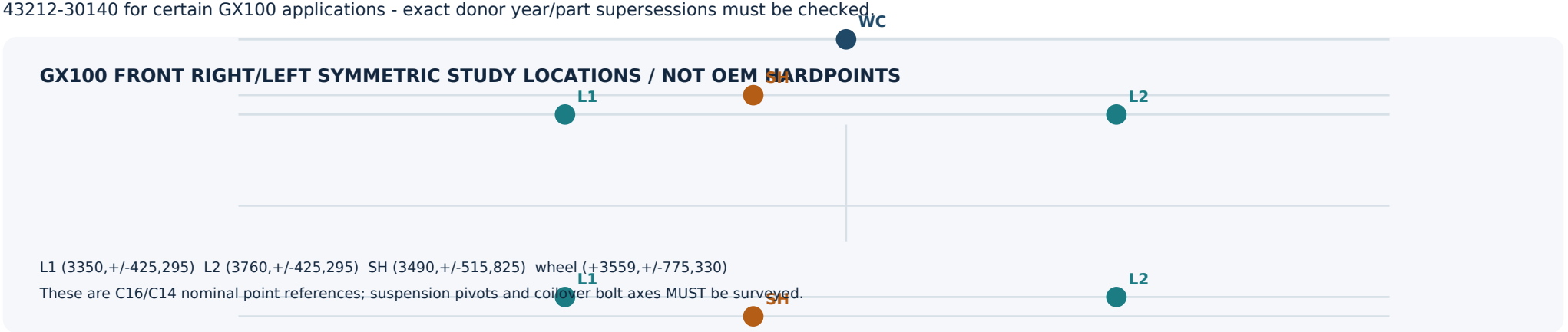
Derived cabin layout: lower outside width 1,070 mm; lower clear inside span 970 mm. The 1,424 mm is a nominal opening near the shoulder zone, not a 1,424-mm floor. C11 contains 21 independently modelled tube solids, but seat, door, pedal, roll-over and seat-belt hardpoints remain undefined.

MANDATORY OCCUPANT HOLD Define actual two-seat installation, seat travel, H-point positions, head clearance, steering/pedals, entry/exit, door cut lines, floor elevation and belt/rollover system before freezing the cabin sill profile.

07 Front GX100 selection and mounting interface

Selected attachment component family

Current primary choice: Toyota Mark II GX100 / X100-family front upright system, double-wishbone, retaining OEM steering knuckle/hub/bearings, correct upper/lower ball joints, and separate lower-arm No. 1 and No. 2 interfaces. Altezza is a studied alternative, not the active default. Relevant catalogue example: right knuckle 43211-30190 and left 43212-30140 for certain GX100 applications - exact donor year/part supersessions must be checked.



Constraint / hardware	Current study value	Status
Front wheel centres	(X +3559, Y +/-775, Z +330) / 1550 track	P - packaging target only
Trial front tyre	660 mm OD x 215 mm wide	P - not an approved tyre
GX100 lower link 1 bracket	(X +3350, Y +/-425, Z +295)	P - C16 proxy
GX100 lower link 2 bracket	(X +3760, Y +/-425, Z +295)	P - C16 proxy
Upper shock/coilover centre	(X +3490, Y +/-515, Z +825)	P - C16 proxy; no OEM fit
Upper arm pivots / steering rack	unknown - measure both sides	M - no trustworthy value
Wheel mounting surfaces / offset	unknown - measure, calculate wheel centrelines	M
Brake and steering clearances	unknown sweep: bump/rebound, lock and full envelope	M/E

08 GX100 brackets / C16 study is not a machining spec

Mount class	C16 nominal geometry	What Codex must do
Lower link No. 1, both sides	One clevis pair per side	Preserve discrete GX100 lower-arm No.1 mount; survey fastener axis, bushing width and clearance.
Lower link No. 2, both sides	One clevis pair per side	Keep independent from No.1; verify caster/camber adjustment provision and link sweep.
Upper wishbone pivots	Two clevis pairs per side	Get the OEM arm pivot-axis positions and measure steering upright and suspension geometry.
Coilover upper	One clevis pair per side	Select damper, eye/stud configuration, joint angles, upper and lower anchor loading.
Concept clevis dimensions	Hole dia 14; ear 6; inner gap 54; bore axis X	P only. Do not preserve blindly or fabricate before exact measured parts and verified load cases.
C16 redesigned support frame	50 retained C15 RHS, 24 new support RHS, 20 clevis ear solids, 4 gusset studies	98 valid geometric STEP solids; topology valid does not equal structurally valid.

Engineering checks for final suspension brackets

- Measure OEM ball centres, WMS, KPI, caster, actual bushing/clevis widths, axes, required motion, end stops and assembly-fastener access.
- Do a virtual kinematics solve for camber curve, caster, toe, bump steer, Ackermann, roll centre, steering lock, wheel sweep, travel and coilover motion ratio.
- Use actual supplier coilover lengths (free/extended/compressed), spring OD, bumpstop and load envelope; C16 has no engineered moving-arm lower damper pickup.
- Select bearing/bushing types, bolts/grades, washer stack, preload/torque and locking plan from actual components; never infer OEM sizing from a trial Ø14 bore.
- Design clevis welds, wall doubler/crush sleeves if structurally necessary, fatigue detail, access windows, drainage and NDT acceptance with a qualified engineer.

Mount sequence

Actual OEM component identification -> dimensional survey -> skeleton model of linkage and WMS -> kinematics -> forces -> mount geometry -> reinforcement/joint analysis -> tool-access and fabrication review -> drill/jig drawings. Never reverse this chain.

09 Rear Probox package / current rear-drive hypothesis

Component / reference	Existing data	Confidence
Donor type	2005-era Toyota Probox, 1NZ-FE petrol, automatic, NCP51V-family believed	P/M - chassis/transmission codes to verify
Engine displacement	1496 cc if correct 1NZ-FE variant	Published related donor info / check car
Original Probox axle module track	1455 front track -> trial new REAR Y +/-727.5	P - specification family, actual module to measure
Powertrain envelope	1100 Y x 850 X x 750 Z nominal	P - crude photo-derived reserved box
Whole removed module width	~1800 without tyres	P - photo estimate
Proposed complete module X space	-650 .. +800	P - C08 reserve, not true engine extent
Old rear frame zone	-791 .. +950	P - C09 outer extents
Rear firewall	+950 X prior	X - incompatible with cockpit aft edge +890
Subframe mounting holes	unknown 3D XYZ and bracket interfaces	M - CRITICAL
Rear MacPherson top mounts	unknown; loose removed struts NOT a template	M - CRITICAL
CV joints / shaft line / outputs	unknown 3D axes and working CV plunge	M
Transaxle code / rotation	U340E/U340F family suggested, actual code not confirmed	M/E - project architecture gate

Rear design approach for Codex

- Use donor OEM subframe, knuckles, hub assemblies, control arms, both struts, transaxle and mounts as independent incoming 1:1 assemblies when measured. Keep structural load paths and load inputs labelled.
- Do not place the strut top coordinates by visually matching loose-strut photographs. Reconstruct the OEM chassis upper anchor geometry from verified body dimensions/physical jig.
- Design a removable engine/subframe service cradle only if the vehicle frame can safely carry all loads and retain critical structures when removed; model fasteners, load transfer and service path.
- Ensure rear powertrain can be safely installed and cooled in its selected orientation. Reversing or reorienting a FWD engine/transaxle is a go/no-go engineering issue, not a framing detail.

10 Front / rear tyres, wheel interfaces and swept body

Constraint	C14/C08 status	Required final input / check
Front track 1550	Nominal centre Y +/-775, GX100 upright study	Design target, revise only after geometry and fender checks.
Rear track 1455	Probox OEM front track as transplanted to rear	Measure actual centres, ride height, camber and axle/subframe width.
Front wheel PCD	GX100 commonly 5 x 114.3	Verify selected hub flange by variant.
Rear wheel PCD	Probox NCP51 commonly 4 x 100	Verify selected hub; consider wheel interchange and spare.
Tyres	660 x 215 front concept; rear specification unknown	Load/speed rating, OD, width, offset and actual tyre section required.
Wheel arch	Starter arch R~444.9 and scaled wheel centre Z~391.5	Does not match front trial tyre centre Z330: reconcile historical silhouette vs hardware.
Steering envelope	C14 reported ~17 mm nominal lateral margin at 30 degrees vs width envelope	Not a wheel/fender dynamic clearance approval. Check full wing shape and suspension movement.
Brake/hub	GX100 front, Probox rear families	Brake master/bias/ABS/parking brake and final masses require system-level engineering.

CAD sweeps to include

- Static wheel/tire at curb and laden height, plus full jounce/rebound and independently full left/right steering sweep; account for tyre growth and tolerances as justified.
- Upper arm, lower arm 1/2, tie rods, steering rack, CV shafts and heat/thermal components along their full motion, including boot/connector clearance.
- Composite fenders and inner wheel wells taken from actual agreed exterior CAD, not just the 2100 mm overall width box.
- Review front/rear unequal PCDs and axle tracks deliberately; do not add wheel adapters or convert hubs without structural and wheel-load justification.

11 Occupant cell, doors, seats and attachments

Hardpoint / attachment	State	Required evidence before bracket design
Seat H-point L / R	M	Human package, adjustable seat rails, head/leg clearance, occupant extremes.
Seat bases and travel tracks	M/E	Actual seat part envelope, clamp geometry, anchored bolts and crash load path.
Seatbelt lap/shoulder anchors	E	Belt geometry and anchor capacity; cannot attach to unsupported RHS wall.
Rollover/overhead protection	E	Protection strategy for two-seat open-top roadster; structure versus period look.
Steering column support	M/E	Wheel, column U-joints, firewall seal, collapsibility, pedal interaction.
Pedal box / master cylinders	M/E	Footwell clearances, front shaft/tunnel, service access and loads.
Door hinges / latch / sill cuts	M/E	Actual door openings, intrusion, egress, hinge reinforcement, anti-sag structure.
Front cowl / windscreen	M/E	Windscreen frame attachment and cowl structure with corrosion/water drainage.
Rear firewall bulkhead	X/E	X+890 vs +950 conflict, fire/heat/impact/odor/noise isolation.
Body mounts	M/E	Composite insert locations, isolating pads, load and washout/drainage.

Door and cabin structural interaction

The open cabin is the weakest torsional region in an unengineered ladder frame. Codex shall study continuous, deep sill load paths, lower diaphragm shear, cowl hoop and rear bulkhead hoop, with localized crash and seat loads. Crossmembers must not obscure occupant feet/under-seat packaging. Belt and seat anchorage designs are safety-critical and require dedicated engineering and certification checks.

12 Mounts, brackets, bolts and weldment engineering

Connection taxonomy

Connection type	CAD / drawing representation	Mandatory check
Main welded RHS node	Actual prepared tube end, weld access surfaces, fit-up gap, orientation; no floating joints.	Shear/moment transfer, fatigue, distortion and accessible inspection.
Suspension clevis	Pair of plates with physical holes, bush/sleeve stack, gap and fastener direction.	Actual OEM bushing axis, bolt spec, wall crushing and peak load.
Engine/subframe mount	OEM subframe bush/bolt references plus engineered cradle structure.	Mount reaction, torque, NVH, fatigue, service and crash.
Seat/belt base	Dedicated plate/backing scheme, fastener axes and designed load paths.	Restraint and seat anchorage compliance, pull-out/tearing.
Body/composite mounting tab	Local steel reinforcement, isolated interface, access and replaceability.	Thermal expansion, water ingress, composite inserts.
Jacking/towing points	Local reinforced structure, labelled loads and travel envelope.	Lifting/towing loads, crush buckling and floor clearance.

Geometry discipline

- Use physically available structural-steel grades with traceable certificates (e.g. ISO/EN/ASTM family selected by supplier and engineer); do not treat the C11 3-mm wall as approved.
- Provide a checked RHS section schedule: h, b, t, tolerance, orientation, steel grade, straight cut/mitre profile, datum points, finished length, quantity and measured mass.
- Specify fillet/butt weld design, symbol standard, size, length, process, quality and NDT plan only after joint review. FreeCAD cut shapes are NOT weld preparations.
- Where tubes cross, distinguish intentional welded coped junction from noncontact clearance; avoid unanalysed internal water traps, unweldable pockets and uncontrolled overlapping material.
- Route brake/fuel/cooling/electrics in protected, clip-supported removable paths through grommeted bulkheads, not through arbitrary thin-wall structural penetrations.
- Document detachable assemblies, captive nuts, threaded inserts and blind fastening options with tool envelopes and replaceability.

13 Structural load cases and required calculation outputs

Case ID	Loading / model required	Deliverable / gate
LC01	Empty / curb mass with real component mass ledger, fluids and occupants	Front/rear wheel static reactions + CoG + ride height.
LC02	Gross design mass, luggage, passengers, fuel, vertical pothole/jounce wheel loads	Global bending, suspension-node displacement, local RHS buckling.
LC03	Asymmetric wheel vertical input / torsion	Torsional stiffness trend, twist angles, local joint stress, full structure support assumptions.
LC04	Peak front/rear braking and acceleration longitudinal reactions	Tyre forces, anti-dive links, rack/jacking brackets; load paths across cabin.
LC05	Cornering / lateral wheel force plus combined longitudinal force	Suspension mount force envelopes, fatigue hotspot screening.
LC06	Steering lock, tie-rod and rack input, bump stops	Rack bracket and suspension hardpoint deflection; kinematics sensitivity.
LC07	Coilover spring/damper maximum and limit/bumpstop loads	Upper/lower coilover mount forces, motion ratio, spring seat fatigue.
LC08	Rear engine/transaxle mount torque and drivetrain traction	Torque reactions, cradle stiffness and exhaust clearance under motion.
LC09	Seat mounts, seats, belts and anchorage / rollover strategy	Dedicated compliance-oriented calculations and qualified review.
LC10	Localized jacking, towing, rear/front impact structural concepts	Defined loads and acceptance framework; no crash claims without test/analysis.
LC11	Thermal: rear exhaust, engine bay, composite and fuel adjacent structure	Heat shields, thermal isolation and monitored verification.

Codex must not invent numerical acceptance limits for torsional stiffness, fatigue endurance, seat belt pull loads or crash criteria. Proposed numerical targets may be researched and documented separately for the actual legal vehicle classification, mass, market and qualified reviewer sign-off.

14 Practical workshop and fabrication constraints

Before cutting any mild-steel RHS

- Procurement: locally available exact RHS/tube dimensions and material certificates; surface finish/galvanizing state, weldability and traceability.
- Manufacturing: saw/laser/plasma capability, machine bed, minimum cope radius, cutting tolerances, weld process, accessibility, fixtures and available lifting gear.
- Jig: master X/Y/Z base, two axle-centre references, centerline, level pads, six or more controlled locating points selected by engineering, diagonal checks and removable clamps.
- Staged construction: primary datum rails and cabin perimeter in jig; tack; remeasure; fit donor proxies and actual components; weld in balanced sequence; inspect distortion before assembly.
- Fastener design: nut access, torque tool clearance, bushing preload at ride height, safety retention and witness marking; no inaccessible blind critical bolts.
- Service: pull engine/transaxle/struts without cutting shell, design removable rear body panels and safe lifting paths.
- Corrosion: seal closed ends by weld or engineered caps; drainage/vent strategy, internal coating, external coat, dissimilar-material composite isolation; no sealed rust pockets.
- Documentation: unique part marks tied to 3D solid, BOM and drawing, joint ID, inspection sign-off and repair/replacement procedure.

Shop deliverables for every approved member and bracket

Part package	Contents
Tube member	Item tag, revision, material certificate ref, cut length convention, centreline/endpoints, 3D solid, end preparation drawing, mass, inspection method.
Laser cut plate	True-scale DXF, plate thickness and grade, hole centres, kerf compensation to shop process, flatness and edge finishing.
Weldment drawing	Overall datums, member IDs, cut orientation, weld details, sequence, inspection zones, distortion/fit-up check.
Jig drawing	Jig origin, nominal pads and stops, clamp clearance, inspection points, tolerances, teardown path.
As-built record	Serial, revision, material heat/batch, weld/test/measurement log, deviations and approvals.

15 CAD master organisation / Codex expected output

Required native model tree

```
Project8C_Master
00_Parameters & Revisions (status per parameter)
01_Body_Reference (scaled starter shell; future validated surfaces)
02_Hardpoint_Skeleton (datums, wheel centres, named MISSING anchors)
03_Donor_Front_GX100 (1:1 measured parts / proxies separately)
04_Donor_Rear_Probox (1:1 measured parts / proxies separately)
05_Chassis_01_Front (frame / weldments / brackets)
06_Chassis_02_Cabin (sills / floor / side / bulkhead)
07_Chassis_03_Rear (frame / cradle / mounts)
08_Interfaces_Global (laps / joints / aligned load paths)
09_Seats_Controls_Belts (envelopes then engineered mounts)
10_Body_Mounts_And_Fluid_Routing
11_Analysis (FEA models, loadcases, reports)
12_Fabrication (cuts / plates / jigs / drawings / QC)
13_Exports (AP242 STEP / DXF / CSV / PDF / JSON)
```

Required geometry pipeline

- Read constraints JSON and source audit; fail loudly on missing mandatory project master parameters.
- Create master axis/scale and transform imported source body loft to master vehicle X. Identify 2100 max width as target, separate from loft-generated central shell.
- Build reference envelopes and hardpoint markers with statuses; show false hardware geometry as distinct bright study layers, never physical OEM solid labels.
- Refactor C16 front, C11 cabin and C09 rear into coherent load-path model; use common datums; resolve +2375..+2551 overlap as engineered physical join; report rear firewall clash.
- Generate tube/plate solids, maintain closed/valid solids, suppress old concept duplicate members, and use per-item metadata with shape provenance.
- Produce exploded/module assembly views, true plan/elevations and full-size cross-sections for engineering review; never label placeholder fasteners as standard OEM sizes.
- Export an unresolved-items report; only produce cutting/weld release packages when user-provided measurements, FEA and qualified approvals are entered.

16 Automated design checks and release logic

Test	Expected programmatic behaviour	On failure
AXIS-01	All parts use rear-axle-origin coordinate system, X front, Y left, Z up.	BLOCK assembly export until transform provenance fixed.
SCALE-02	Body reference uses 2100/1770 exactly; donor solids remain at 1:1 scale.	BLOCK mixing of body and donor scales.
ZONE-03	All three modules in one global spatial frame; physical splice geometry described.	REPORT dangling/open junctions; block release.
SKIN-04	Compare frame geometry to validated body skin and mounting keep-out volumes.	REPORT unsupported skin intersections / unknown fenders.
TIRE-05	Swept tyre/arm/rack envelope and wheel offsets evaluated under approved travel.	BLOCK clearance approval while measurements missing.
MOUNT-06	No safety-critical drilled hole with only P status.	HARD BLOCK machining/weld release.
RHS-07	Tube cross-sections valid, length positive, correct thickness, no duplicate bodies.	BLOCK STEP if invalid; flag intentional contacts separately.
WELD-08	Every structural endpoint assigned to node/joint and inspectable weld plan.	REPORT disjoint members and inaccessible welds.
FEA-09	Loads, supports, mass, mesh sensitivity, stress and deflection evidence documented.	BLOCK engineering release if absent/failed.
ASBUILT-10	Drawing cut lengths, 3D solids, BOM, jig datums and part IDs cross-checked.	BLOCK fabrication release on mismatch.

File names Codex should produce

```
Project_8C_Master_Chassis.FCStd
Project_8C_Master_Chassis_Assembly.step
Project_8C_Chassis_01_Front.step
Project_8C_Chassis_02_Cabin.step
Project_8C_Chassis_03_Rear.step
Project_8C_Datum_and_Hardpoint_Registry.csv
Project_8C_Weldment_Member_Schedule.csv
Project_8C_Bracket_Drilling_HOLD.csv
Project_8C_Conflict_Resolution_Log.pdf
Project_8C_Concept_Drawing_Set.pdf
Project_8C_Design_Validation_Report.pdf
```

17 Measured-item acquisition and gates

Gate	Missing input / decision	Who / evidence	Design impact
G01	Confirm Probox VIN family/year, actual engine + gearbox code, drive layout.	Donor owner / service labels & OEM manual	Rear transaxle go/no-go.
G02	Measure Probox subframe bolts, strut upper mounts and output/wheel axes.	Physical 3D survey with datums	Rear frame + strut tops.
G03	Select exact GX100 upright / hub / lower links 1&2 / upper arm.	OEM parts + measured assemblies	All front link geometry.
G04	Confirm front steering rack/tie rods, wheel offset, brake rotor / calipers.	Matched donor/hardware survey	Bump steer, clearance, brakes.
G05	Select front coilovers and lower-arm mounting solutions.	Supplier dimensions and loads	Mount hole/plate design.
G06	Resolve scaled body fender+wheel-arch CAD and actual wheel package.	Exterior surface + tyre drawings	Wheel travel, body/rail clearance.
G07	Choose physical seats, cabin interior package and belt/rollover strategy.	Measured components + qualified review	Sill width/height, attachments.
G08	Resolve rear firewall at +890/+950 with engine to +800.	Packaging sections + fire strategy	Cabin length and bulkhead.
G09	Mass budget, materials, service access, protected plumbing.	Calculation + supplier data	Load cases and tube sizing.
G10	Complete FEA, fatigue, weld procedures, jig and inspection.	Independent engineer + workshop	Permits fabrication release.
G11	Confirm BRTA approval classification and required standards.	Official authority / qualified specialist	Road-legal release criteria.

Interim permitted progress

Codex may develop validated concept geometry, investigate manufacturable joint alternatives, generate collision reports and build a complete editable CAD tree with HOLD markers. Any physical cutting, drilling, structural weld instructions or road-safety claims remain blocked on the relevant gates.

18 Codex execution charter / paste into the engineering run

Task: Rebuild Project 8C as one parameterised, fabrication-oriented welded steel chassis, organized into front, cabin and rear assemblies. Use this PDF and companion JSON/CSV as the binding project design brief; the files and prior STEP models are references, not authority for unmeasured Toyota interfaces.

Mandatory work order

- Read the dossier and write an input acceptance report listing all current project-fixed, derived, provisional, missing and conflicting constraints.
- Construct the master geometry and import any accessible C11/C16/rear STEP files with scale/axis verification. If source CAD files are unavailable, create clean proxy reference models without pretending they are the originals.
- Issue at least two structural architecture alternatives (e.g. deep sill frame with cross-braced floor vs reinforced RHS semi-spaceframe), and select one based on load paths, jig access, cabin ingress and donor fit - not visual complexity.
- Create the parametric skeletal frame. Bring C16 lower No.1/No.2 and upper arm brackets into robust front load paths; no drilled release features until actual component surveys.
- Rework cabin sills, crossmembers and rear bulkhead to occupant / body fit. Show and solve front overlap X2375..2551; quarantine the firewall X890/950 conflict.
- Develop rear subframe cradle with measured-point placeholders and explicit FWD transaxle-feasibility gate; keep powertrain removal path and rear crash structure.
- Make production-oriented member segmentation, square/mitre cuts, weld joint volumes, bolt access and jig points; output the items listed in Section 16.
- Perform geometric and interference checks and setup FEA/kinematic studies with transparent load assumptions. Publish every missing engineering calculation as NOT VERIFIED.
- Give a changed-parameters log, design review drawings, open issues list and proposed next physical measurements. Never claim released-for-manufacture without external sign-off.

SUCCESS DEFINITION	A coherent, editable and well-documented full-length chassis design study with proper donor placeholders, front/cabin/rear load paths, and transparent fabrication engineering gates - not merely a visually impressive STEP compound.

19 Technical provenance and source index

Ref	Project reference / evidence	Used for
S01	Project 8C preliminary engineering plan (21 pp, 11 Oct 2026).	System architecture, hold points, QA, workshop and legal gate.
S02	Project 8C Planning.txt / uniform scaling C03 text.	2100/1770, 6110, 3559, 1780 design targets.
S03	Body starter: Project_8C_Body_Starter.FCMacro (ZIP).	Provisional loft stations and original s coordinate direction.
S04	Project_8C_C10_Cabin_Dimensions_From_Scaled_Silhouette.pdf and CSV.	C10 cockpit bounds, body stations and 60-mm clash.
S05	Project_8C_C11_Cabin_RHS_Frame_Concept.pdf + STEP + member CSV.	C11 cabin envelope and 21-member trial arrangement.
S06	Project_8C_Chassis_Three_Sections_Cabin_Envelope_Concept.pdf.	Prior 3-way split and legacy rear firewall plane.
S07	Clarify ChS.txt / C08 powertrain photo dimensional study.	1NZ-FE Probox module, 1455 track and rear layout unknowns.
S08	Project C12 GX100/Altezza upright selection notes.	Part families, sample knuckle IDs, wheel PCD issue.
S09	Project C14 GX100 front suspension concept notes.	Front track, tyres, preliminary wheel centres / hardpoint caveats.
S10	Project C15 integrated front chassis concept notes.	Front span +2375..4545, nose +4805 and rail integration.
S11	Project C16 GX100 front chassis mounts engineering review, 4 pp.	C16 mounts and 98-solid concept, trial 6/54/14 clevis values.

Independently cross-checked part catalogue family (not mounting geometry)

Toyota Mark II GX100 front axle arm/steering knuckle catalogue: https://toyota.epc-data.com/mark_ii/gx100/80249/chassis/4802/48068M/
GX100 RH knuckle example part reference: https://toyota.epc-data.com/mark_ii/gx100/80243/chassis/4802/43211/
GX100 genuine-parts diagram incl. No.1 and No.2 lower arms: https://partsouq.com/en/catalog/genuine/diagram?c=Toyota
No catalogue source above provides a validated as-installed set of 3D mounting coordinates for the donor in this project.

20 Release matrix and explicit responsibility boundaries

Deliverable level	Allowed now	Not allowed without evidence
RESEARCH / CONCEPT	Compare frame routes, show proxy donor parts, schematic geometry and alternate joints.	Claim OEM parts fit because CAD bracket appears aligned.
PARAMETRIC CAD	Build STEP/FreeCAD master, 2D schematics, status-tagged skeleton, geometry QA.	Assign false precision to strut, suspension, seat-belt or weld hardpoints.
ENGINEERING REVIEW	Mass/axle loads, detailed survey, preliminary FEA and kinematics with documented assumptions.	Represent simulated or theoretical adequacy as certification.
PROTOTYPE / JIG DESIGN	Engineer validated member cuts, bracket holes, WPS and weld sequence.	Fabricate when materials, weld, fit-up and safety approvals are incomplete.
RELEASE FOR MANUFACTURE	Signed BOM, cut files, verified FEA/kinematics, 3D datum inspection and tolerances.	Authorise road use; roadworthiness requires separate testing and legal approval.

Shortlist of top integration risks

- R01: FWD transaxle repositioning for rear-drive installation - orientation, final-drive rotation, lubrication, shifter, axle and cooling must be verified.
- R02: Cabin firewall physical plane (+890 / +950) and engine forward envelope to +800 make an extremely tight packaging region.
- R03: Scaled body loft does not include validated fender geometry; wheel/tyre position is not proven.
- R04: GX100 upper/lower link pivots, steering gear, bushing axes and coilover dimensions remain physical-survey items.
- R05: Cabin open-roof torsion, side intrusion, rollover and seat/belt anchorage cannot be solved from C11 tubes alone.
- R06: There are no approved steel material, wall thickness, tube joint, weld, load limit, regulatory or jig tolerances.

End of C17 design brief. Companion JSON and CSV files are intended as the input contract and register for the initial Codex modelling run. Parameter changes must be written back to a controlled revision log with source and approval.